Customer Behavioral Segmentation For Targeted Marketing

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1. Problem Description

- The Customer Segmentation System is a Machine Learning based project developed using Python and K-Means Clustering algorithm to analyze customer purchasing behavior and divide customers into different groups according to their similarities.
- The project helps businesses identify valuable customer groups based on factors like annual income, spending score, age, and purchase behavior. By using clustering techniques, companies can improve marketing strategies, customer satisfaction, product recommendations, and business decision-making.
- This project performs data preprocessing, feature scaling, clustering, visualization, and evaluation of customer groups using various Machine Learning libraries.

1.1 Dataset Description

The dataset used in this project contains customer shopping and spending information.

Features of Dataset:

- Customer ID
- Gender
- Age
- Annual Income
- Spending Score
- Purchase Frequency

The dataset is cleaned and preprocessed before applying the clustering algorithm. Categorical values are converted into numerical form and feature scaling is applied for better clustering performance.

1.2 Flow Diagram

Fig.1 Sample flow diagram.

2.Modules for Project Implementation

2.1 Modules Used

1. Pandas
Used for data manipulation and analysis.
Functions Used:

- pd.read_csv() → To load dataset
- df.head() → To display first rows
- df.info() → To display dataset information
- df.describe() → Statistical summary

2.
NumPy
Used for numerical operations. Functions
Used:

- np.array()
- np.mean()

3. Matplotlib
Used for plotting graphs and cluster visualization. Functions
Used:

- plt.scatter()
- plt.plot()
- plt.xlabel()
- plt.ylabel()
- plt.show()
-

4. Seaborn
Used for advanced data visualization.
Functions Used:

- sns.heatmap()
- sns.pairplot()

5. Scikit-learn
Used for Machine Learning implementation. Modules
Used:

- KMeans
- StandardScaler
- silhouette_score
- wcss

2.2 Platform Used:

Jupyter Notebook
Used for implementing Machine Learning algorithms, visualization, and code execution.

Visual Studio Code
Used for writing, debugging, and testing Python code.

Python
Programming language used for data preprocessing, clustering, and visualization

3. Machine Learning Model/ Searching Technique Used(Describe each ML model/ searching technique used)

K-Means Clustering

K-Means Clustering is an unsupervised Machine Learning algorithm used to divide data into different clusters based on similarities between data points. The algorithm works in the following steps:

1. Select number of clusters (K)
2. Randomly initialize centroids
3. Assign data points to nearest centroid
4. Update centroids
5. Repeat until convergence

The project uses the Elbow Method and Silhouette Score to determine the optimal number of clusters. Advantages:

- Simple and efficient
- Works well with large datasets
- Easy visualization of customer groups

Limitations:

- Sensitive to outliers
- Requires predefined K value

4. Evaluation metrics used

1. WCSS (Within Cluster Sum of Squares)
WCSS measures the distance between data points and their cluster centroid.
Formula:

$$WCSS = \sum (x_i - c_i)^2$$

Where:

- x_i= data point
- c_i= centroid

Lower WCSS indicates better clustering.

2. Silhouette Score
Silhouette Score measures how well data points fit within their clusters.
Formula:

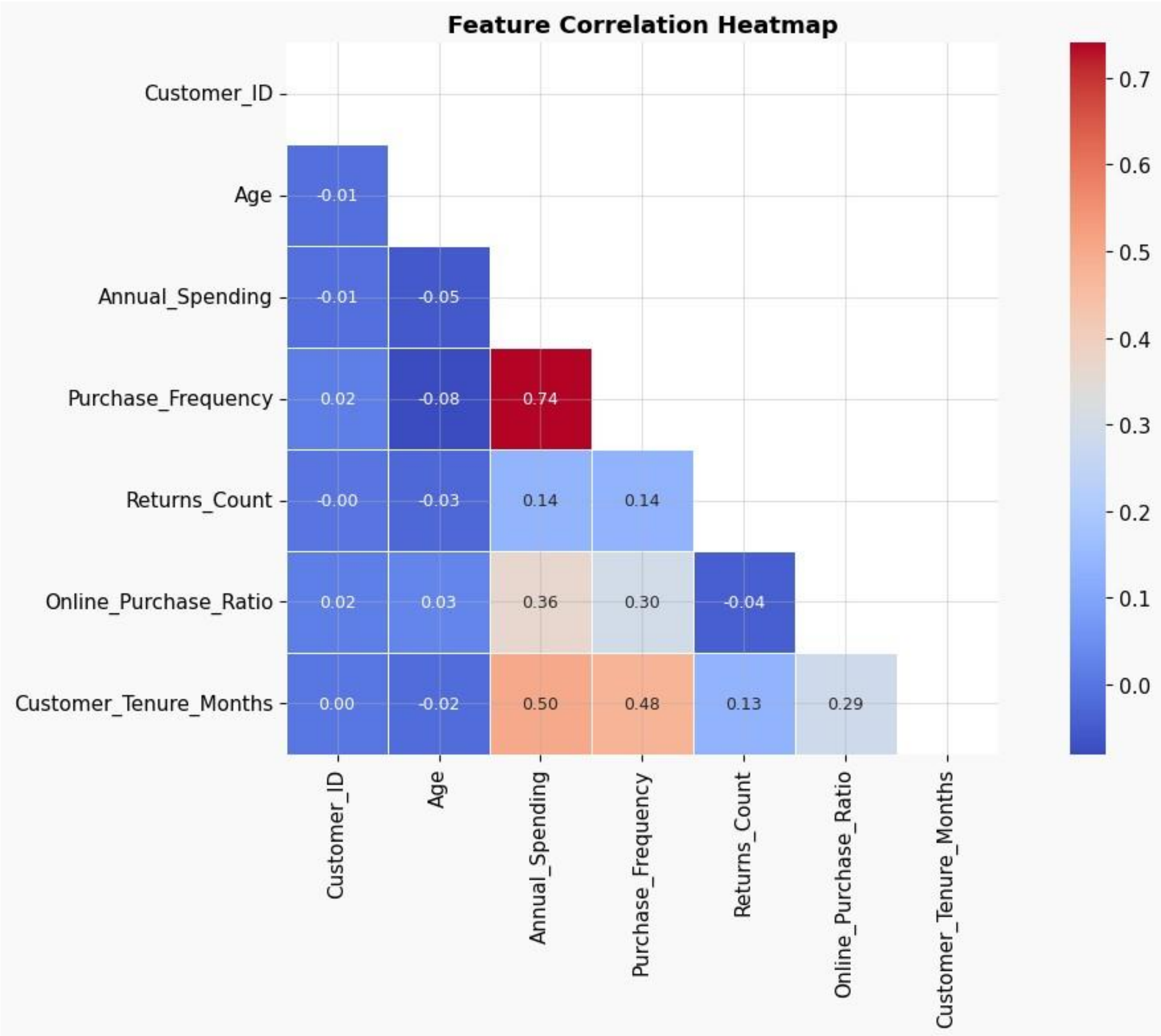
$$S = \frac{b - a}{max(a, b)}$$

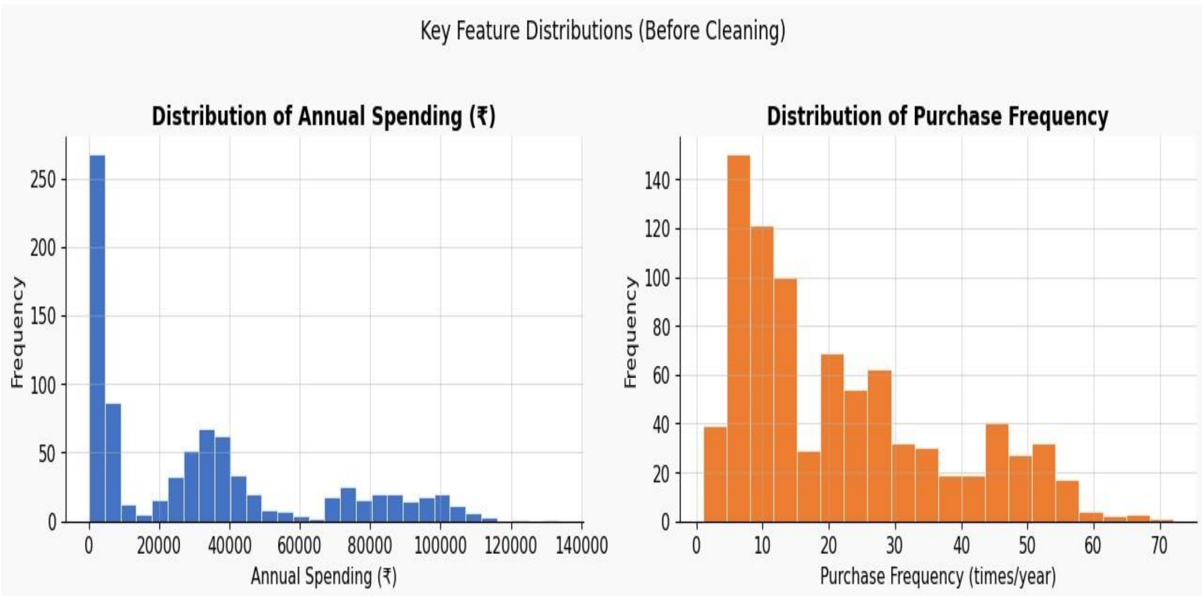
Where:

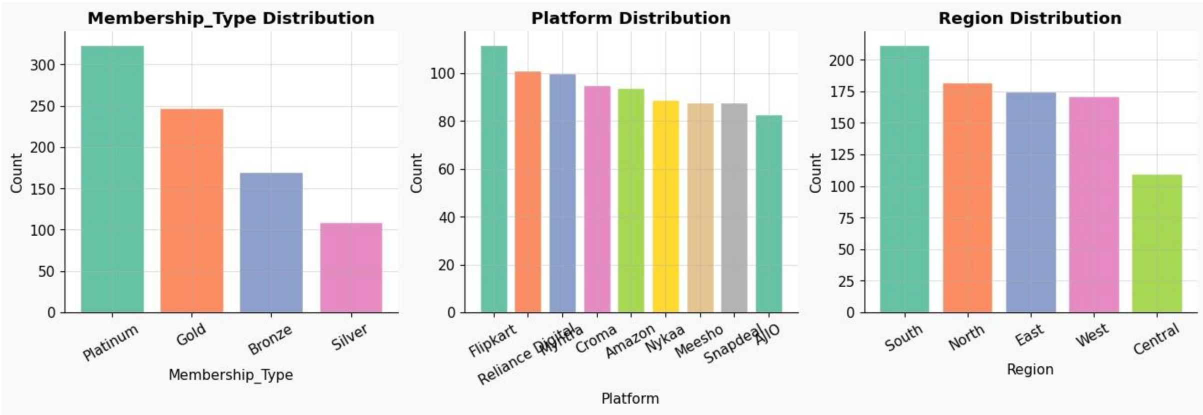
- a= average intra-cluster distance
- b= average nearest-cluster distance Value ranges from -1 to 1.

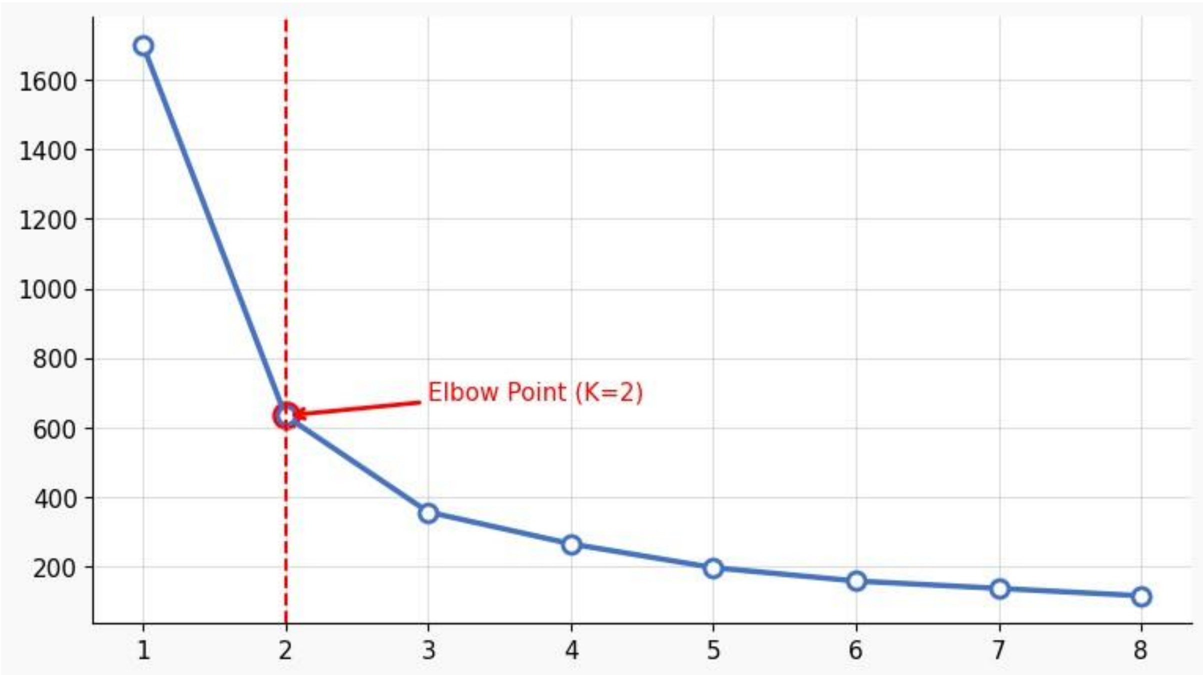
Higher value indicates better clustering.

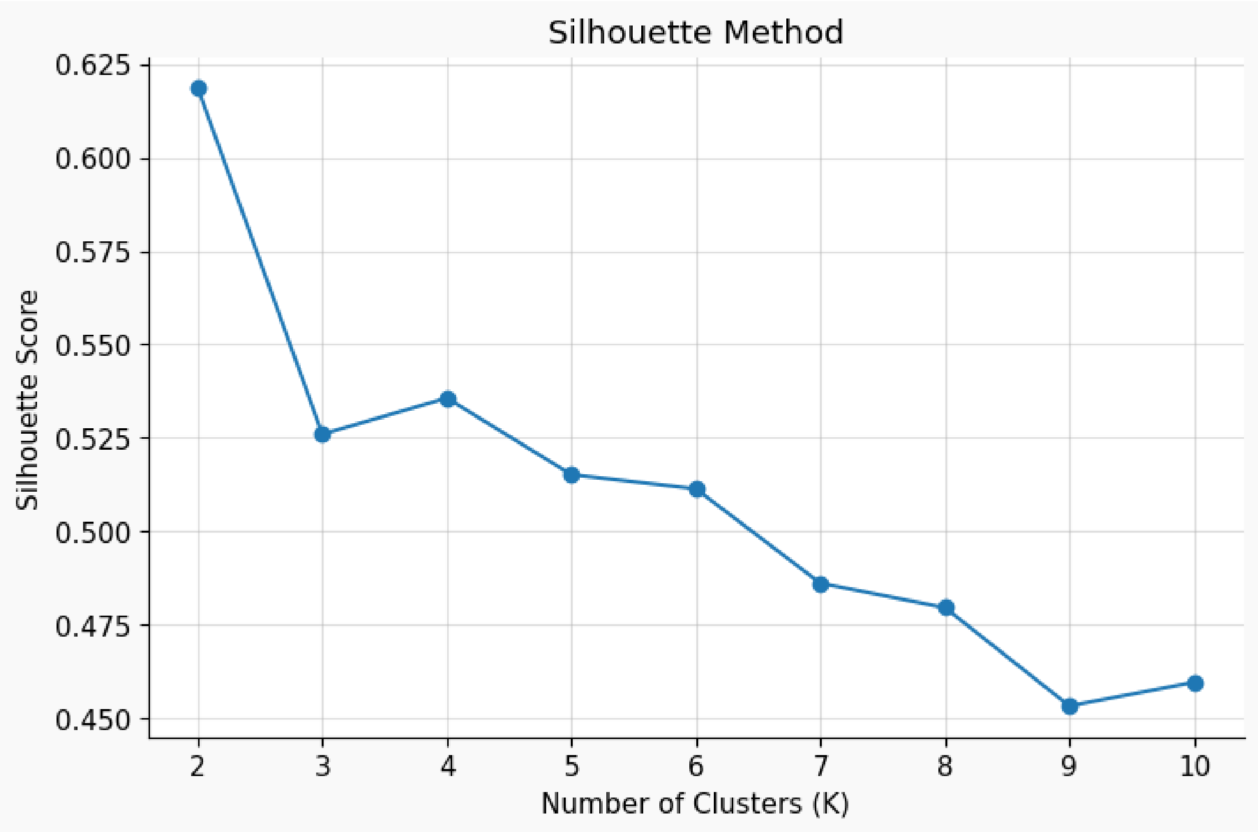
5. Results and Visualizations

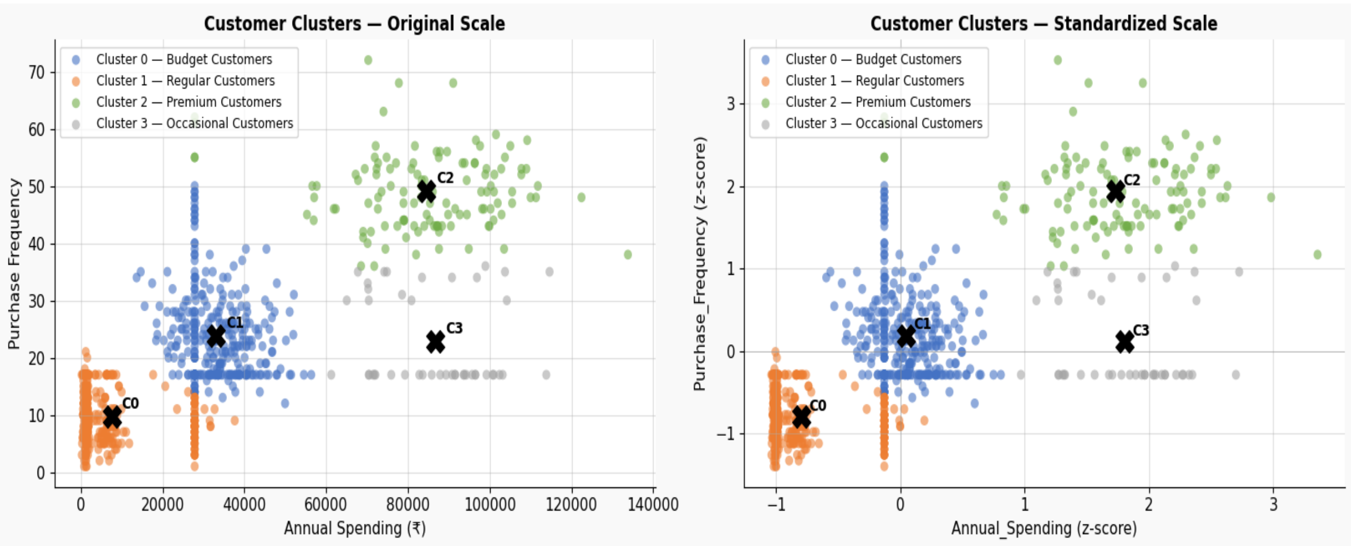





Membership_Type Distribution
Count
Platinum
Gold
Bronze
Silver
Membership_Type
Platform Distribution
Count
Flipkart
Reliance Digital
Myntra
Croma
Amazon
Nykaa
Meesho
Snapdeal
AJIO
Platform
Region Distribution
Count
South
North
East
West
Central
Region


Elbow Point (K=2)


Silhouette Method
Silhouette Score
Number of Clusters (K)
0.625
0.600
0.575
0.550
0.525
0.500
0.475
0.450
2
3
4
5
6
7
8
9
10


Customer Clusters — Original Scale
Cluster 0 — Budget Customers
Cluster 1 — Regular Customers
Cluster 2 — Premium Customers
Cluster 3 — Occasional Customers
C0
C1
C2
C3
Purchase Frequency
Annual Spending (₹)
Customer Clusters — Standardized Scale
Purchase_Frequency (z-score)
Annual_Spending (z-score)

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CUSTOMER PERSONA PROFILES

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Budget Customers (Cluster 0) — 313 customers (36.8%)

Avg Annual Spending : ₹33,040
Avg Purchase Freq : 23.9 times/year
Avg Returns Count : 2.07
Avg Online Ratio : 46.09%
Avg Tenure (months) : 36.0

Regular Customers (Cluster 1) — 374 customers (44.0%)

Avg Annual Spending : ₹7,508
Avg Purchase Freq : 9.7 times/year
Avg Returns Count : 0.93
Avg Online Ratio : 40.92%
Avg Tenure (months) : 24.0

Premium Customers (Cluster 2) — 119 customers (14.0%)

Avg Annual Spending : ₹84,459
Avg Purchase Freq : 49.1 times/year
Avg Returns Count : 1.39
Avg Online Ratio : 58.03%
Avg Tenure (months) : 47.2

Occasional Customers (Cluster 3) — 44 customers (5.2%)

Avg Annual Spending : ₹86,631
Avg Purchase Freq : 22.8 times/year
Avg Returns Count : 1.18
Avg Online Ratio : 55.10%
Avg Tenure (months) : 46.6

6. Conclusion and Future Scope:

The project successfully implemented Customer Segmentation using K-Means Clustering algorithm. Different customer groups were identified based on purchasing behavior and spending patterns.
The clustering results help businesses improve targeted marketing strategies and customer relationship management.

Future Scope:

- Use larger real-world datasets
- Implement DBSCAN and Hierarchical Clustering
- Deploy project as web application
- Add real-time customer analysis

7. References:

Books:

1. Python Machine Learning – Sebastian Raschka 2.
Hands-On Machine Learning – Aurélien Géron
3.

Online Resources:

- GeeksforGeeks
- Scikit-learn Documentation
- Kaggle
- Stack Overflow